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HGC^3AE^2 at the Degraded Edge (v1.1 errata)

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v1.1-preprint

How to cite

Justin H. Kuiper, CISSP (2026). *HGC^3AE^2 at the Degraded Edge (v1.1 errata)* (v1.1-preprint). Non Sequitur Publishing. https://nonsequitur.tech/white-papers/hgc3ae2-at-the-degraded-edge/

HGC³AE² at the Degraded Edge (v1.1 errata)

Contents

Abstract

The HGC³AE² framework specifies the structural commitments — human-governed authority, curated context, agentically engineered execution, runtime verification — that distinguish trustworthy operation from capable but ungoverned operation in agentic systems.[1] Those commitments were established in conditions that the cloud-native deployment substrate satisfies by default: continuous connectivity to authoritative governance, accessible curated-context tiers, observability planes that aggregate continuously, and adversarial pressure as-

[1] Justin H. Kuiper, CISSP, *HGC³AE² Framework* (v1.0-preprint, 2026), zenodo.org/records/19869285.

sumed to be commercial rather than operational. The Edge Degraded Operations substrate (EDO-P1) satisfies none of these by default.[2] This paper develops HGC^{3}AE^2 as it must operate under DDIL conditions: how Human-governance is preserved through pre-authorized envelopes that survive disconnection; how Curated Context survives partition; how Cybersecurity is enforced under adversarial pressure on every layer; how Continuity becomes the architecture's primary design property; how Coherence is maintained across reconciliation cycles; how Agentic Engineering operates within disconnection-tolerant envelopes; and how Execution is verified runtime when the verification machinery itself may be partially disconnected. The argument is that HGC^{3}AE^2 is preserved under DDIL conditions, not weakened — the architectural commitments survive translation to the tactical substrate, provided the translation is done explicitly rather than assumed.

How to cite

Justin H. Kuiper, CISSP (2026). *HGC^{3}AE^2 at the Degraded Edge* (v1.0-preprint-draft). Non Sequitur Publishing.

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1. Introduction

The HGC^{3}AE^2 framework was developed against the cloud-native deployment substrate: continuous connectivity, accessible centralized governance, observable persistence, commercial adversarial pressure.[3] The framework's structural commitments — what humans must do, what curation must preserve, what agentic execution must satisfy, how runtime verification closes the loop — were specified in terms of operational primitives that the cloud-native substrate supports natively.

The Edge Degraded Operations substrate (EDO-P1) supports those primitives differently. Connectivity is intermittent; centralized governance may be unreachable; persistence is anchored at the edge node with reconciliation across boundaries; adversarial pressure is operational rather than commercial.[4] HGC^{3}AE^2's commitments do not become irrelevant under these conditions; they become **load-bearing in a different way**. The H of HGC^{3}AE^2 cannot rely on continuous human consultation; it must rely on pre-authorized envelopes that survive disconnection. The G of curated context cannot rely on a centralized tier; it must operate over local curated state with explicit reconciliation discipline. The C^1 of cybersecurity must assume hostile substrate rather than benign substrate.

This paper translates each letter of HGC^{3}AE^2 into its degraded-edge form. The argument is that the framework's commitments are preserved under DDIL conditions; the translation is the architectural work that the Edge Degraded Operations Undecalogy carries out across its eleven papers. §2 names five failure modes that occur when cloud-native HGC^{3}AE^2 is deployed onto the tactical substrate without translation. §3 specifies the letter-by-letter translation. §4 treats the translation as a continuous process. §5 binds the translated framework to Skipjack. §6 returns the analysis to HGC^{3}AE^2 and frames the remainder of the Undecalogy.

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[2] EDO-P1 *Tactical Substrate* promotion (v1.0-preprint, in concurrent drafting; yks-build#103).
[3] Justin H. Kuiper, CISSP, *HGC^{3}AE^2 Framework* (v1.0-preprint, 2026), zenodo.org/records/19869285.
[4] EDO-P1 *Tactical Substrate* promotion (v1.0-preprint, in concurrent drafting; yks-build#103).

2. Failure Modes of Untranslated HGC³AE² on the Tactical Substrate

Five failure modes recur when cloud-native HGC³AE² implementations are deployed onto the tactical substrate without explicit translation.

Halt-on-disconnect. A cloud-native implementation of H — continuous human governance consultation — halts when governance is unreachable. The agentic system loses operational capability, often precisely when operational capability is most needed.[5]

Stale-curation drift. Local curated context drifts from authoritative state during disconnection. When connectivity returns, reconciliation surfaces divergence the system has been operating against. Without explicit drift handling, the divergence either accumulates silently or forces costly rollback.[6]

Privileged-channel collapse. Cloud-native security assumes privileged channels (KMS, IAM, attestation services) are continuously reachable. Under disconnection, these channels collapse; security state cannot be verified; the system either operates against stale verification (risky) or halts (also risky).[7]

Observability blackout. Continuous observability assumes continuous connectivity to the centralized telemetry plane. Under disconnection, telemetry buffers locally; when capacity is constrained, buffers exhaust; observability data is lost. Operator awareness of the system's state during disconnection becomes a reconstructed estimate at best.[8]

Reconciliation gaps. Reconnection does not automatically merge the disconnected substrate's state with the authoritative source. Without explicit reconciliation discipline, the substrate either over-writes authoritative state (loss), is over-written by it (loss), or operates in inconsistent dual-state (drift).

The unified problem: **cloud-native HGC³AE² implementations fail structurally on the tactical substrate**. The remainder of this paper specifies the translation that preserves the framework.

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3. HGC³AE² Letter-by-Letter at the Degraded Edge

3.1 H — Human-governed via Pre-Authorized Envelopes

Human governance under DDIL is exercised through **pre-authorized envelopes**: explicit, persistent specifications of what the agentic system may do, against what context, with what authority, while disconnected from continuous human consultation. The envelope is authored by the governance layer when connectivity is available; it persists locally on the substrate; the substrate operates inside the envelope without further authorization until reconnection.[9]

The envelope is not a static permission set. It is a structured specification with explicit conditions for proceeding, conditions for halting, conditions for escalation, and conditions for

[5] Vinton G. Cerf et al., "Delay-Tolerant Networking Architecture," RFC 4838, IETF, 2007.

[6] Marc Shapiro et al., "Conflict-Free Replicated Data Types," *SSS '11*, 2011, https://doi.org/10.1007/978-3-642-24550-3_29.

[7] Ali Humayed et al., "Cyber-Physical Systems Security — A Survey," *IEEE Internet of Things Journal* 4, no. 6 (2017): 1802–1831, https://doi.org/10.1109/JIOT.2017.2703172.

[8] Saksham Tushar et al., "Observability in Microservices: A Systematic Literature Review," *Information and Software Technology* 158 (2023): 107173, https://doi.org/10.1016/j.infsof.2023.107173.

[9] Justin H. Kuiper, CISSP, *Alistair Prime in a Box* (v1.0-preprint, 2026), zenodo.org/records/19869307.

fallback. The envelope encodes what humans authorized; the substrate enforces the encoding.

3.2 G — Curated Context Under Partition

Local curated context survives partition by being explicitly local-authoritative within the substrate during disconnection, with explicit reconciliation discipline upon reconnection. The CRDT primitives developed in EDO-P4 are the operational instruments: distributed-state types that admit concurrent updates with deterministic merge semantics.[10][11] Curation discipline is preserved through local-authoritative status rather than through continuous central authority.

3.3 C¹ — Cybersecurity Under Hostile Substrate

Security under the degraded edge assumes hostile substrate: cryptographic privacy, integrity attestation, and signed governance events are first-class concerns at every layer. Privileged-channel design must include local-fallback modes for the disconnected case: local key material with explicit rotation discipline; local attestation chains with reconciliation-time verification; signed governance event logs that the substrate can verify offline.[12][13]

3.4 C² — Continuity as Primary Design Property

Continuity becomes the architecture's primary design property under DDIL. Workloads survive disconnection by operating against pre-authorized envelopes; persistence survives connectivity loss by anchoring on local substrate with explicit reconciliation; governance state survives partition by carrying forward the most recently authoritative envelope until reconciliation updates it.

3.5 C³ — Coherence Across Reconciliation Cycles

Coherence is maintained not by continuous availability but by **reconciliation cycles**. Each cycle merges the substrate's accumulated state with the authoritative source's state using deterministic merge semantics. Coherence is therefore episodic-but-correct, not continuous; the architectural commitment is that the reconciliation merge is deterministic and verifiable.[14]

3.6 A — Agentically Engineered Within Disconnection-Tolerant Envelopes

Agentic execution operates within disconnection-tolerant envelopes specified by the governance layer at pre-authorization time. The agentic system does not extend its envelope under disconnection; it operates inside it. Capability without permission is the cloud-native failure mode HGC³AE² names; capability without reachability is the DDIL variation, and the architectural response is the same — the envelope, pre-authorized and locally enforced.

[10] Marc Shapiro et al., "Conflict-Free Replicated Data Types," *SSS '11*, 2011, https://doi.org/10.1007/978-3-642-24550-3_29.

[11] EDO-P4 *State Coherence Problem* promotion (in concurrent drafting; yks-build#96).

[12] Bob Callaway et al., "Sigstore: Software Signing for Everybody," *CCS '22*, 2022, https://doi.org/10.1145/3548606.3560596.

[13] Jingbin Liu, Yu Qin, Dengguo Feng, "RIM-EI: Toward Trusted Attestation of Persistent State in Trusted Execution Environments," *IEEE Transactions on Dependable and Secure Computing* 19, no. 4 (2022): 2580–2596, https://doi.org/10.1109/TDSC.2021.3060127.

[14] Werner Vogels, "Eventually Consistent," *Communications of the ACM* 52, no. 1 (2009): 40–44, https://doi.org/10.1145/1435417.1435432.

3.7 E — Runtime Verification Under Partial Disconnection

Runtime verification under DDIL operates locally: the substrate verifies its own runtime against pre-authorized invariants without continuous external consultation. Verification machinery must itself survive disconnection; the substrate must be able to verify its operation even when external verification services are unreachable.

3.8 The Exponent — The Disconnected-Reconcile-Adjust Loop

The HGC³AE² exponent closes through the disconnected-reconcile-adjust cycle (EDO-P1 §4). Telemetry accumulated during disconnection feeds the next reconciliation; reconciliation feeds governance-layer envelope adjustment; the adjusted envelope propagates to the substrate for the next disconnected operating cycle. The loop closes asynchronously; the closure is preserved.

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4. The Translation as a Continuous Process

The letter-by-letter translation is not a one-time architectural exercise. The substrate's operating conditions evolve: connectivity profiles change, adversarial pressure intensifies, capacity oscillations shift. The translation must be revisited at sprint-aligned Skipjack cadence, with envelope adjustments propagating the revised translation to the substrate.

The three-phase cycle from EDO-P1 (disconnected operation → reconciliation → adjustment) is the operational instantiation of the translation's continuity.

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5. Skipjack Coupling Under Translation

The Skipjack mechanisms operate under the translated HGC³AE² with the adaptations introduced in EDO-P1 §5.[15][16] **Context integrity** holds locally during disconnection and is verified during reconciliation. **Structured routing** operates within the pre-authorized envelope. **HITL control points** fire when envelope boundaries are approached; under disconnection, control points become *halt-and-await-reconnection* events. **Iterative feedback** closes asynchronously across reconciliation cycles.

The mechanisms preserve their commitments under translation. The Skipjack Protocol is one of the architectural commitments HGC³AE² depends on, and its translation to the tactical substrate is what makes the framework operational at the degraded edge.

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6. Implications and Conclusion

The implications of translating HGC³AE² to the degraded edge extend to architecture (pre-authorized envelopes become first-class artifacts), tooling (CRDT and WAL primitives, attestation chains, local verification machinery require dedicated investment), operational doctrine (DDIL ops is a discipline), and adoption sequencing (organizations move to the degraded edge in phases, not all at once).

[15] Justin H. Kuiper, CISSP, *Skipjack Protocol* (v1.0-preprint, 2026), zenodo.org/records/19869293.
[16] EDO-P1 *Tactical Substrate* promotion (v1.0-preprint, in concurrent drafting; yks-build#103).

The core claim: **HGC^{3}AE^2's structural commitments are preserved under DDIL conditions through explicit translation; the translation is the architectural work that the Edge Degraded Operations Undecalogy carries out across its eleven papers**.

Returned to HGC^{3}AE^2 letter by letter — this paper's §3 is the return; the rest of the Undecalogy develops each translated letter in operational depth.[17][18][19][20]

References

— end v1.0-preprint-draft —

Drafting state: v1.0-preprint draft (promotion from v0.1-seed-rev3). PR count: 12 inline (Cerf RFC · Shapiro SSS · Humayed IoT-J · Tushar IST · Vogels CACM · Callaway CCS · Liu TDSC · Shi IoT-J · Terry SOSP · Fall SIGCOMM · Kelly DSN + supporting). §3 §6 load-bearing reconciliation across all 8 letters of HGC^{3}AE^2 for degraded-edge translation.

Appendix — v1.1 Errata Addendum

Added 2026-05-17 per Round-2 EDITORIAL-STAMP HOLD-substance verdict. The v1.0-preprint remains canonical at its original DOI; this v1.1 supplements with the corrections below.

§X. v1.1 Errata — Round-2 Substance Corrections

§X.1. Why this addendum exists

EDO-P2 v1.0-preprint shipped on 2026-04-30 at DOI 10.5281/zenodo.20180148 and has been reader-visible at `nonsequitur.tech/pubs/white-papers/hgc3ae2-at-the-degraded-edge/` since that date. The paper translates the seven letters and exponent of HGC^{3}AE^2 to the DDIL tactical substrate, establishing the architectural frame that the remaining Edge Degraded Operations Undecalogy papers develop in operational depth.

On 2026-05-17, the post-publish Round-2 editorial review (Edna, second-pass under §1.5 EDITORIAL-STAMP doctrine, parent thread YKS-Spine-Binder#355) returned a verdict of **HOLD-substance**. The review identified two substance defects and one citation-hygiene defect. The substance defects are not cosmetic: §4 lacks the operational weight required of a standalone section in this paper's load-bearing structure, and §6 violates the HGC^{3}AE^2 return discipline that the framework's own publication doctrine requires of concluding sections. Both defects shipped in the reader-visible artifact.

This addendum names each defect, states the v1.0 text being corrected, and provides the canonical correction. The v1.0 DOI is not revoked; the v1.0 text remains as published. This addendum ships as a companion artifact under a separate Zenodo record, and v1.1 of the paper incorporates the corrections specified here into the body text.

[17] EDO-P3 *When the Link Is the Variable* promotion (in concurrent drafting; yks-build#95).
[18] EDO-P4 *State Coherence Problem* promotion (in concurrent drafting; yks-build#96).
[19] EDO-P5 *Orchestration Problem* promotion (in concurrent drafting; yks-build#97).
[20] EDO-P11 *Deployment Problem* CAPSTONE promotion (in concurrent drafting; yks-build#110).

§X.2. Defect: §4 substance-thin — bridge paragraph inflated to section status

Round-2 finding: "§4 Translation as Continuous Process — Substance-thin. Two paragraphs. States that the translation must be revisited as conditions evolve and names the three-phase cycle from EDO-P1 as the instantiation. Does not animate: no trigger criteria for when revision fires, no example of what a revised translation looks like, no operational distinction between a translation revision and an envelope adjustment (which §3.1 already covers). This section reads as a bridge paragraph inflated to section status. HOLD trigger #1." (Round-2 stamp, 2026-05-17, §4 audit)

v1.0 said: "The letter-by-letter translation is not a one-time architectural exercise. The substrate's operating conditions evolve: connectivity profiles change, adversarial pressure intensifies, capacity oscillations shift. The translation must be revisited at sprint-aligned Skipjack cadence, with envelope adjustments propagating the revised translation to the substrate. The three-phase cycle from EDO-P1 (disconnected operation → reconciliation → adjustment) is the operational instantiation of the translation's continuity." (§4, complete text)

Canonical correction: §4 must be expanded to carry substance commensurate with its section-level status, or demoted to a bridging paragraph within §3. The expanded version must include:

1. **Trigger criteria.** Specify what conditions cause the translation to be revisited: connectivity-profile change beyond a named threshold, adversarial-pressure escalation surfaced through reconciliation-cycle telemetry, capacity loss that invalidates a pre-authorized envelope's resource assumptions, or governance-layer directive. The trigger is not calendar-driven; it is condition-driven, with a sprint-cadence review as the backstop.
2. **Operational example.** At minimum one concrete scenario: e.g., an envelope authorized under a connectivity profile of "90-minute reconnection windows every 6 hours" encounters a substrate shift to "no reconnection for 48 hours." The translation revision re-derives the continuity commitment (C^2) under the revised profile, adjusts the envelope's halt-and-await-reconnection thresholds, and propagates via the next reconciliation cycle.
3. **Distinction from envelope adjustment.** §3.1 specifies envelope adjustment as a governance-layer act within the current translation. §4's scope is the translation itself — the mapping from framework letter to substrate primitive — not the envelope content. The section must name this distinction explicitly: envelope adjustment operates within a translation; translation revision changes the mapping the envelopes are built on.

Why: A section that does not animate its own claims to operational specificity falls below the substance floor for a load-bearing paper in the Undecalogy. The three requirements above are the minimum; the author may develop further. [FORWARD — EDO-P1 §4 three-phase cycle, yks-build#103, pending v1.0]

§X.3. Defect: §6 violates HGC³AE² return discipline

Round-2 finding: "§6 Implications and Conclusion — the HGC³AE² return discipline is violated. §6 organizes implications by domain (architecture, tooling, doctrine, adoption) rather than walking H-G-C^1-C^2-C^3-A-E-E individually. The paper acknowledges at §6 line 'this paper's §3 is the return' — but doctrine requires the conclusion to independently walk the letters, not delegate the return to an earlier section. HOLD trigger #2." (Round-2 stamp, 2026-05-17, §6 audit)

v1.0 said: "The implications of translating HGC³AE² to the degraded edge extend to architecture (pre-authorized envelopes become first-class artifacts), tooling (CRDT and WAL primitives, attestation chains, local verification machinery require dedicated investment), operational doctrine (DDIL ops is a discipline), and adoption sequencing (organizations move to the degraded edge in phases, not all at once). […] Returned to HGC^3AE^2 letter by letter — this paper's §3 is the return; the rest of the Undecalogy develops each translated letter in operational depth." (§6, selected text)

Canonical correction: §6 must walk the letters independently in its concluding structure. The domain-organized implications (architecture, tooling, doctrine, adoption) may remain as a framing paragraph, but the section's primary structure must be a letter-by-letter return: H → what this paper established for human governance at the degraded edge and what remains open; G → same for curated context under partition; C^1 → cybersecurity under hostile substrate; C^2 → continuity as primary design property; C^3 → coherence across reconciliation cycles; A → agentic execution within disconnection-tolerant envelopes; E → runtime verification under partial disconnection; and the exponent → the disconnected-reconcile-adjust loop. Each letter's return should be 2–4 sentences: what §3 established, what the downstream Undecalogy paper develops, and what the reader should hold as the standing commitment. The delegation pattern ("this paper's §3 is the return") is not compliant; §6 must independently close the loop.

Why: The HGC^3AE^2 return discipline exists because a paper that invokes the framework's letter structure in its body must verify, in its conclusion, that every letter received substantive treatment. Delegating the return to §3 makes §6 structurally redundant